

Demo: Modeling Mass Diffusion in Materials Science Applications

“Bytes & Bonds”

Hosted by the Wang Materials Group

Today's plan

What is Materials Science and Engineering?

Examples

Materials Science Tetrahedron

What is diffusion?

Where is diffusion important?

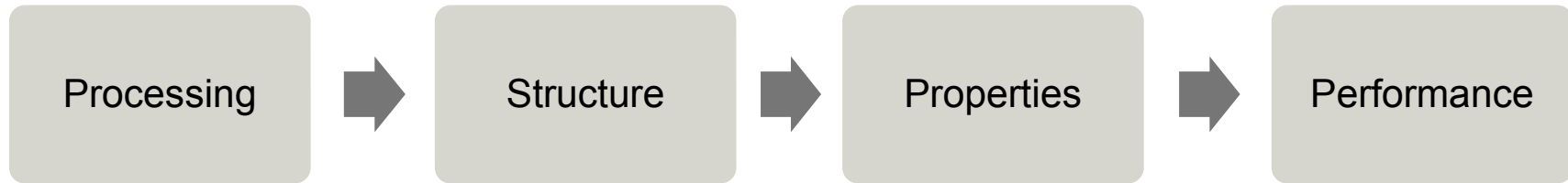
What is computation and simulation?

Random Walk Diffusion

Using simulations to model diffusion

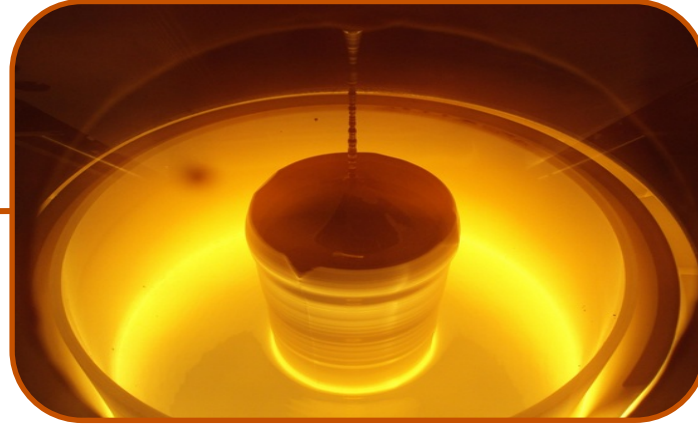
What is materials science and engineering?

Discovering, understanding, and designing materials (often times solids) through the study and application of processing-structure-property-performance relationships



What is materials science and engineering?

How is the material actually made in a lab/factory or adapted from nature?



Processing



Structure

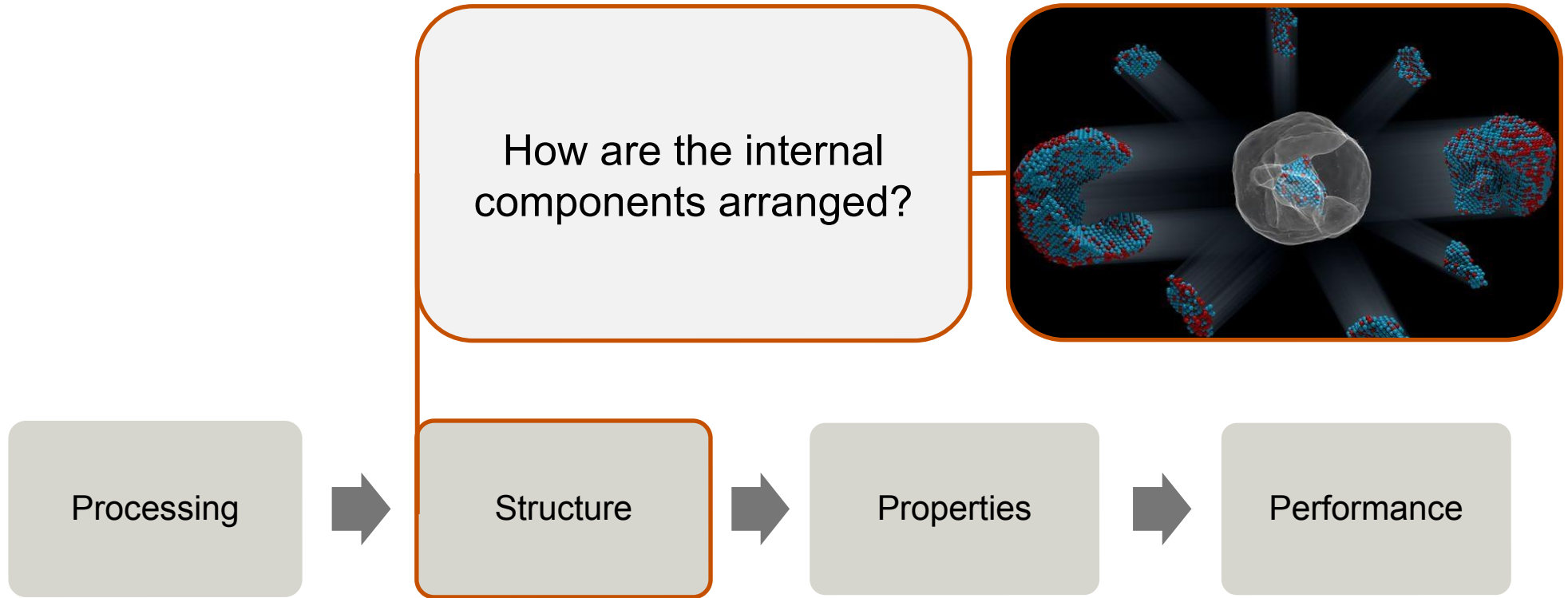


Properties

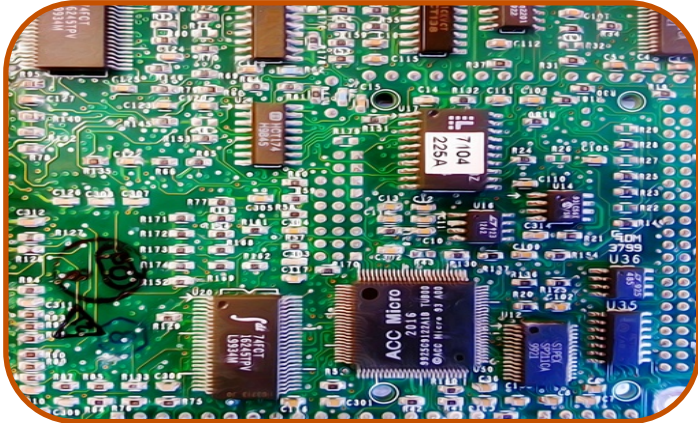


Performance

What is materials science and engineering?



What is materials science and engineering?



How does it respond to an external stimulus?

Mechanical

Electrical

Thermal

Magnetic

Optical

Chemical

Processing



Structure

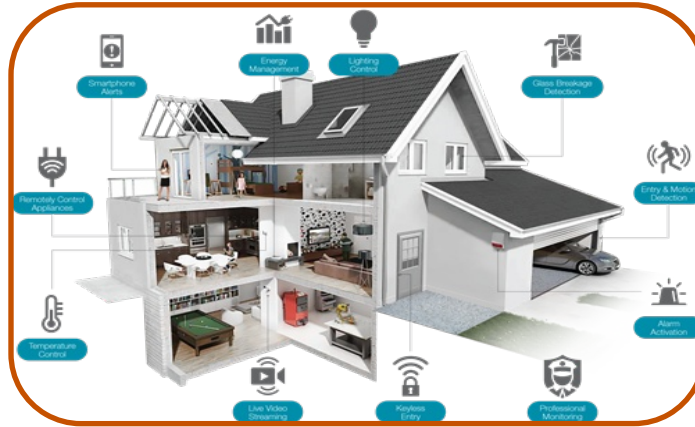


Properties



Performance

What is materials science and engineering?



How does the material help the user/world?

Processing



Structure

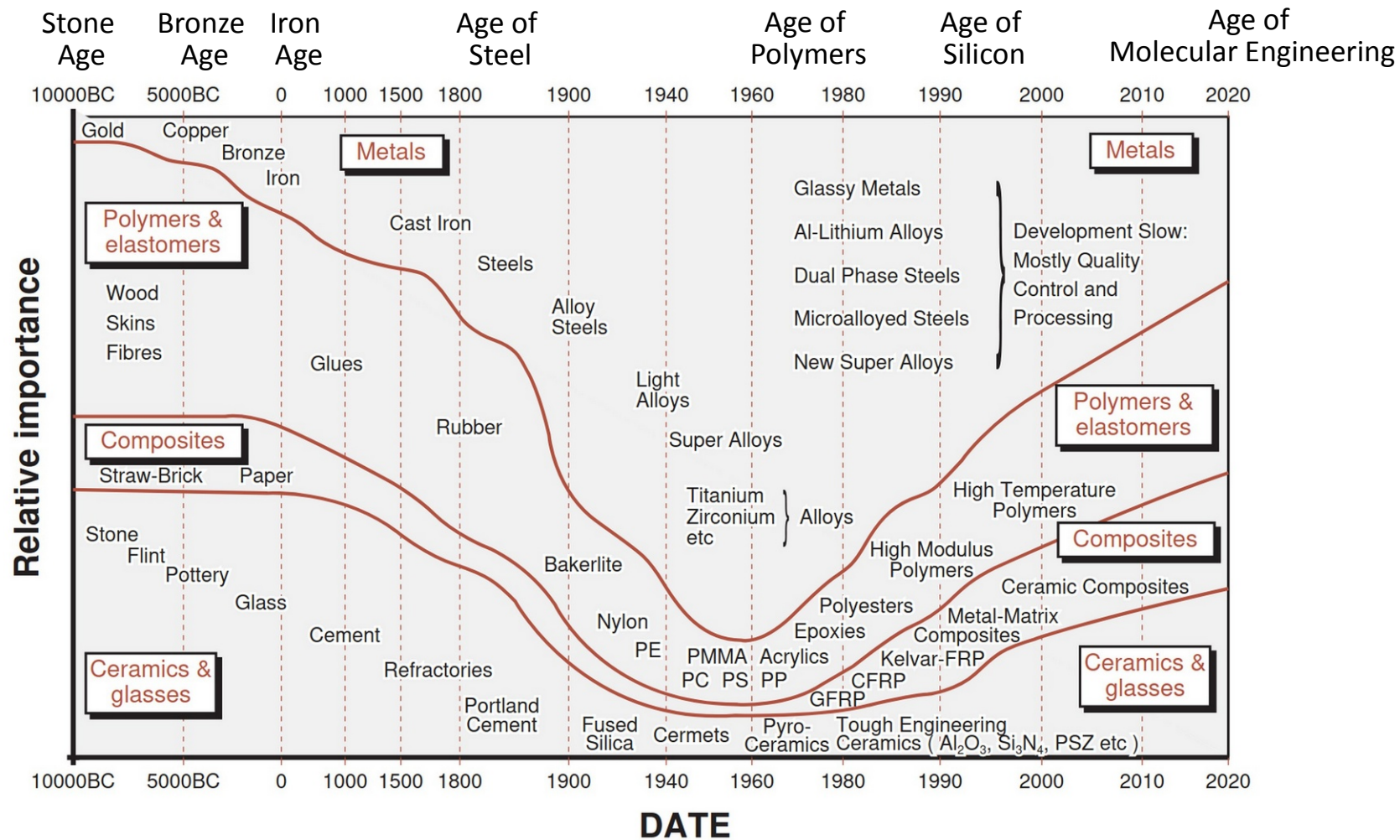


Properties



Performance

Why study materials? (Think-pair-share)



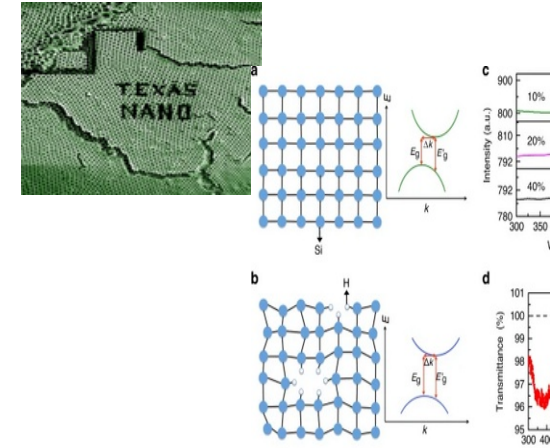
Examples of Materials Science and/or Engineering



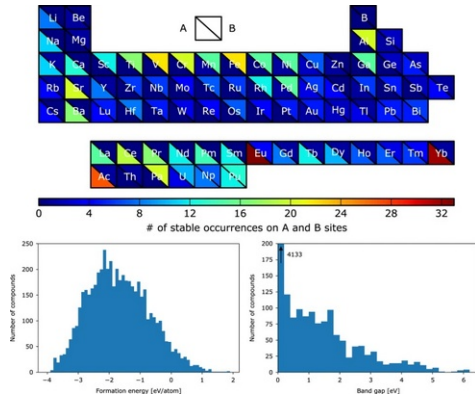
3D printing soft bio-polymers
Nelson Lab, University of Washington



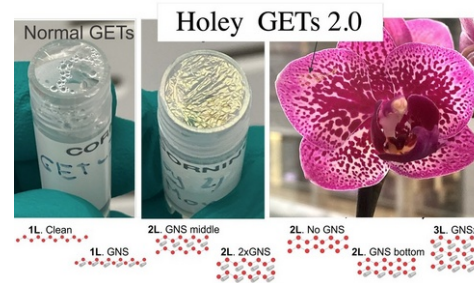
Using viruses as templates to
make more efficient solar cells
Belcher Lab, MIT



Using defects in nanoparticles for high-efficiency, low-loss nanophotonics
Korgel & Zheng Research Groups, UT Austin



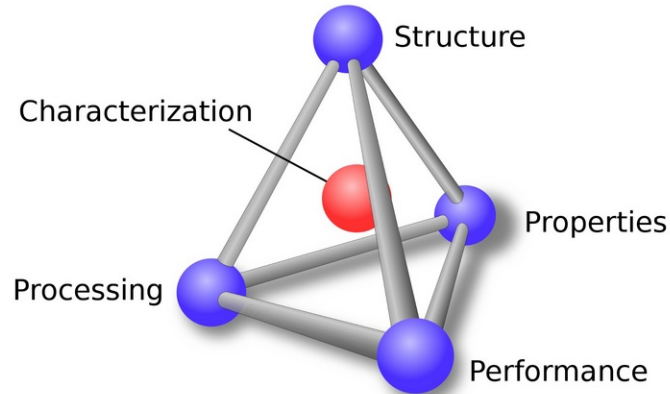
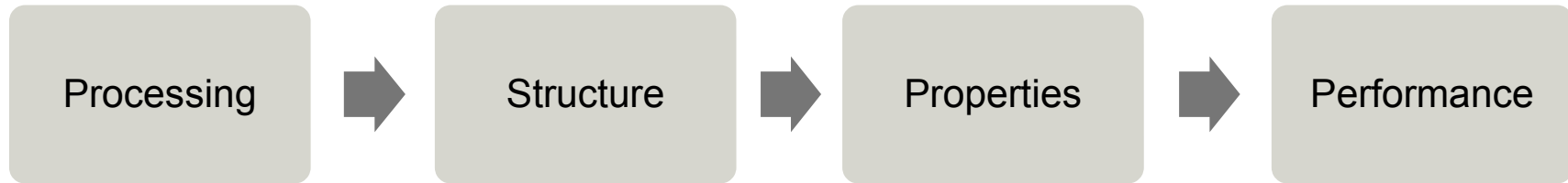
Using supercomputers to discover
new and ideal oxides
Wolverton Research Group, Northwestern



Breathable graphene tatoos
for personalized healthcare
Akinwande Nano Research Group, UT Austin

What is materials science and engineering?

Discovering, understanding, and designing materials (often times solids) through the study and application of processing-structure-property-performance relationships



What is diffusion?



<https://www.youtube.com/watch?v=UlnubVuJvZM>

In-person activity: Human Diffusion

Step 0: Find a space at least twice as large as the space occupied by participants.

What happens to the distribution of the people in the room?

Step 1: Pack all participants to one side of the room as close as possible. Each participant faces a different direction.

Step 2: Each participant begins shuffling (slowly and carefully!) in the direction they are facing.

Step 3: If a participant runs into another object (e.g., wall or other participant), s/he moves in another random direction.

Step 4: Repeat for steps 2 and 3 many times.

What is diffusion? The effect of temperature

What happens to the distribution of the ink in the flask?



*not strictly mass diffusion

Everyday examples of diffusion



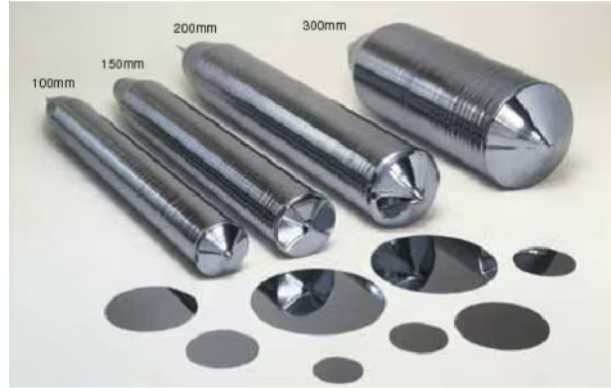
CO₂ (gas) dissolved
in flavored liquid



Aerosols, air fresheners, perfumes



Air pollutants



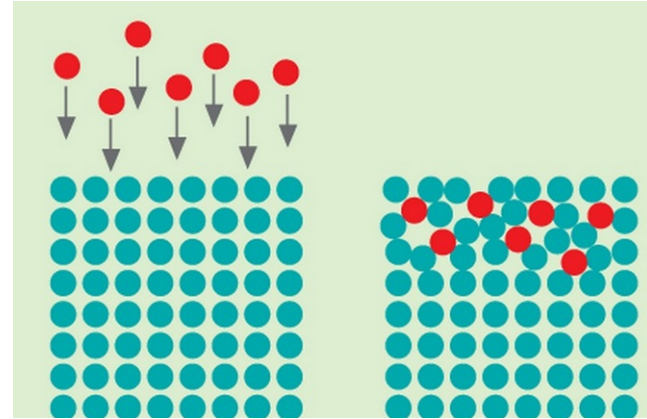
1	H Hydrogen																2	He Helium																																																								
3	4																5	B Boron	6	C Carbon	7	N Nitrogen	8	O Oxygen	9	F Fluorine	10	Ne Neon																																														
11	12																13	Al Aluminum	14	Si Silicon	15	P Phosphorus	16	S Sulfur	17	Cl Chlorine	18	Ar Argon																																														
19	20																21	K Potassium	22	Ca Calcium	23	Sc Scandium	24	Ti Titanium	25	V Vanadium	26	Cr Chromium	27	Mn Manganese	28	Fe Iron	29	Co Cobalt	30	Ni Nickel	31	Cu Copper	32	Zn Zinc	33	Ga Gallium	34	Ge Germanium	35	As Arsenic	36	Se Selenium	37	Br Bromine	38	Kr Krypton																						
39	40																41	Rb Rubidium	42	Sr Strontium	43	Y Yttrium	44	Zr Zirconium	45	Nb Niobium	46	Mo Molybdenum	47	Tc Technetium	48	Ru Ruthenium	49	Rh Rhodium	50	Pd Palladium	51	Ag Silver	52	Cd Cadmium	53	In Indium	54	Sn Tin	55	Sb Antimony	56	Te Tellurium	57	I Iodine	58	Xe Xenon																						
59	60																61	Cs Cesium	62	Ba Barium	63	La Lanthanum	64	Hf Hafnium	65	Ta Tantalum	66	W Tungsten	67	Re Rhenium	68	Os Osmium	69	Ir Iridium	70	Pt Platinum	71	Au Gold	72	Hg Mercury	73	Tl Thallium	74	Pb Lead	75	Bi Bismuth	76	Po Polonium	77	At Astatine	78	Rn Radon																						
79	80																81	Fr Francium	82	Ra Radium	83	Ac Actinium	84	Rf Rutherfordium	85	Db Dubnium	86	Sg Seaborgium	87	Bh Bohrium	88	Hs Hassium	89	Mt Meitnerium	90	U Uranium	91	Pa Protactinium	92	Th Thorium	93	Uu Ununium	94	Uub Unbibium	95	Uut Ununtrium	96	Uuq Ununquadium	97	Uuh Ununhexium	98	Uus Ununseptium	99	Uuo Ununoctium	100	Uu11 Ununundecium	101	Uu12 Ununbium	102	Uu13 Ununtrium	103	Uu14 Ununquadium	104	Uu15 Ununpentium	105	Uu16 Ununhexium	106	Uu17 Ununseptium	107	Uu18 Ununoctium	108	Uu19 Ununennium	109	Uu20 Unbium
109	110																111	Cn Copernicium	112	Nh Nihonium	113	Fl Flerovium	114	Mc Moscovium	115	Lv Livermorium	116	Ts Tennessine	117	Ug Oganesson																																												
118	119																120	Uue Ununennium	121	Uub Unbium	122	Uut Ununtrium	123	Uuq Ununquadium	124	Uuh Ununhexium	125	Uus Ununseptium	126	Uuo Ununoctium	127	Uu11 Ununundecium	128	Uu12 Ununbium	129	Uu13 Ununtrium	130	Uu14 Ununquadium	131	Uu15 Ununpentium	132	Uu16 Ununhexium	133	Uu17 Ununseptium	134	Uu18 Ununoctium	135	Uu19 Ununennium	136	Uu20 Unbium																								
136	137																138	Ce Cerium	139	Pr Praseodymium	140	Nd Neodymium	141	Pm Promethium	142	Sm Samarium	143	Eu Europium	144	Gd Gadolinium	145	Tb Terbium	146	Dy Dysprosium	147	Ho Holmium	148	Er Erbium	149	Tm Thulium	150	Yb Ytterbium	151	Lu Lutetium																														
152	153																154	Th Thorium	155	Pa Protactinium	156	U Uranium	157	Np Neptunium	158	Pu Plutonium	159	Am Americium	160	Cm Curium	161	Bk Berkelium	162	Cf Californium	163	Es Einsteinium	164	Fm Fermium	165	Md Mendelevium	166	No Nobelium	167	Lr Lawrencium																														

- Alkali metals ● Alkaline earth metals ● Transition metals ● Post-transition metals
● Metalloids ● Reactive nonmetals ● Noble gases ● Lanthanides
● Actinides ● Unknown properties

Silicon Wafer Processing: Example of Diffusion in Materials Science

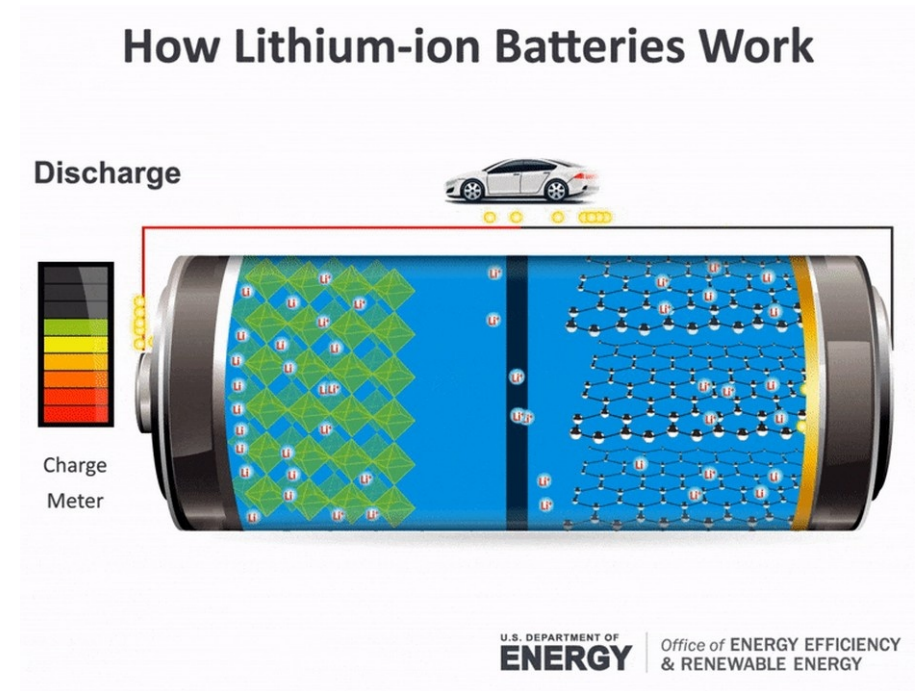
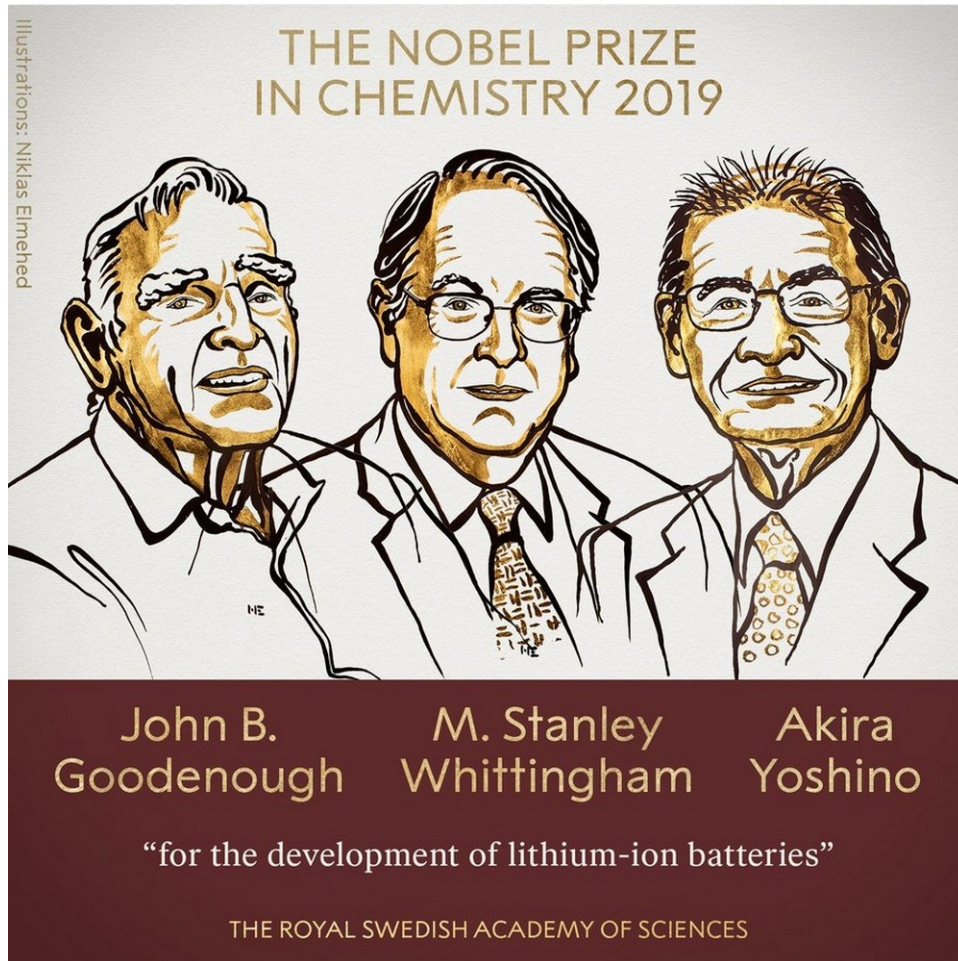
Intentional incorporation of impurities (e.g., boron, phosphorous)

Step 1: Steady-state gas diffusion
or ion implantation



Step 2: Drive-in process (higher temperature)
→ uniform distribution of impurities

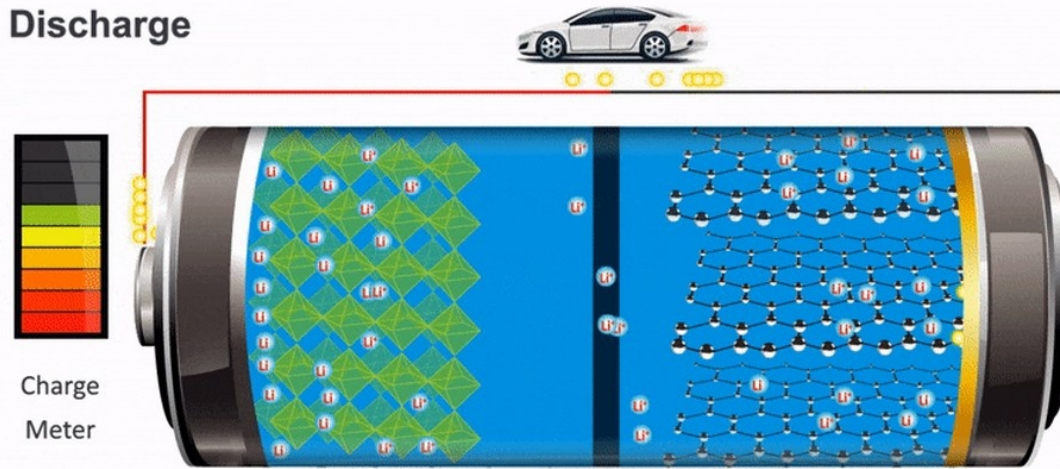
What is the materials science behind the Li-ion battery winning the 2019 Nobel prize in Chemistry?



Diffusion of Lithium ions in a battery

How Lithium-ion Batteries Work

Discharge

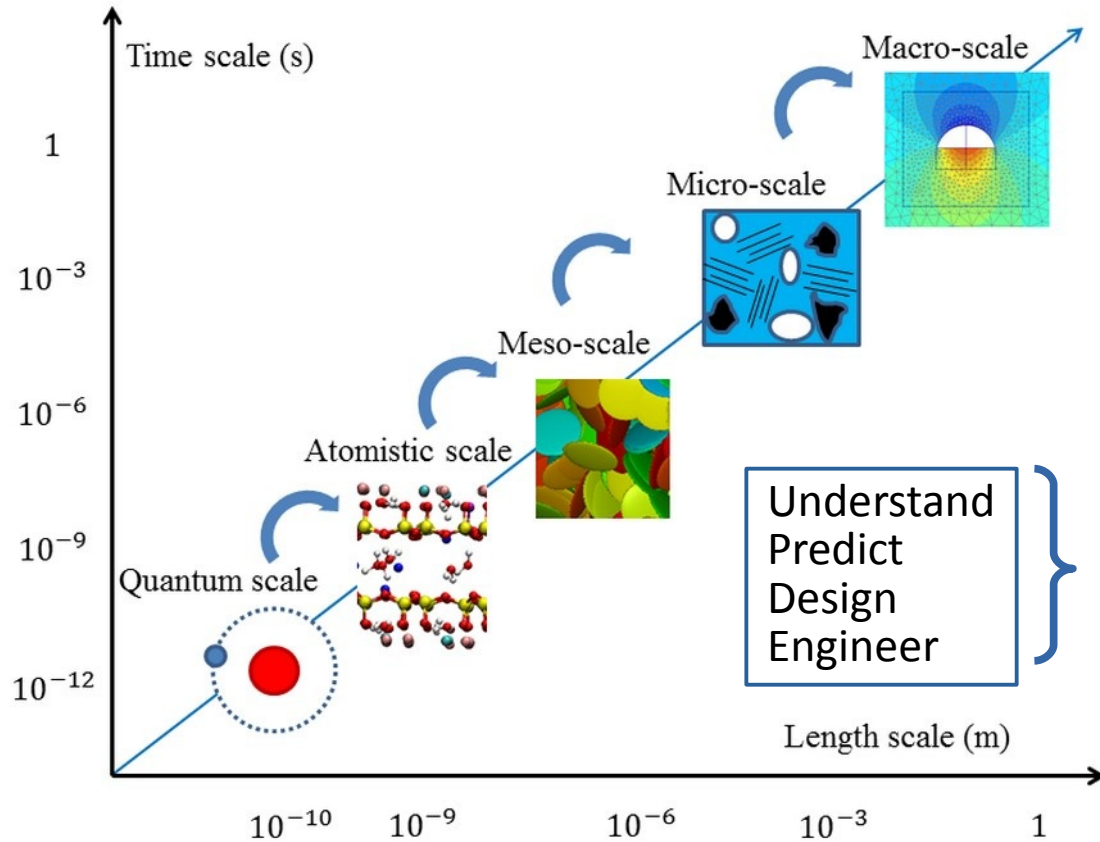


U.S. DEPARTMENT OF
ENERGY

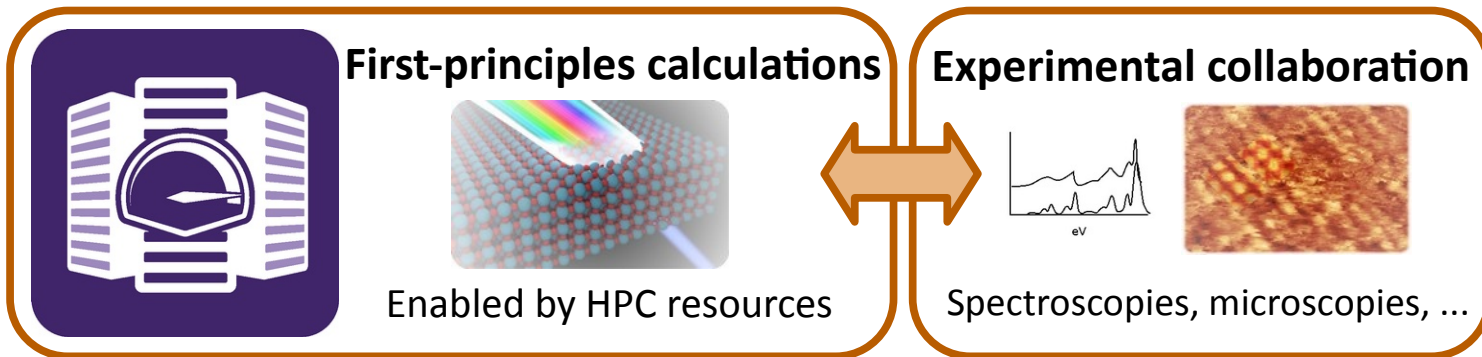
Office of ENERGY EFFICIENCY
& RENEWABLE ENERGY

<https://www.energy.gov/science/doe-explainsbatteries>

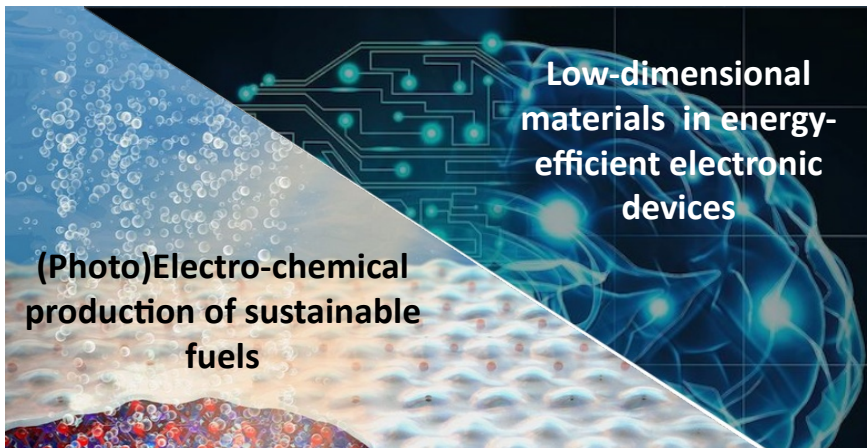
What is Modeling and Simulation in Materials Science?



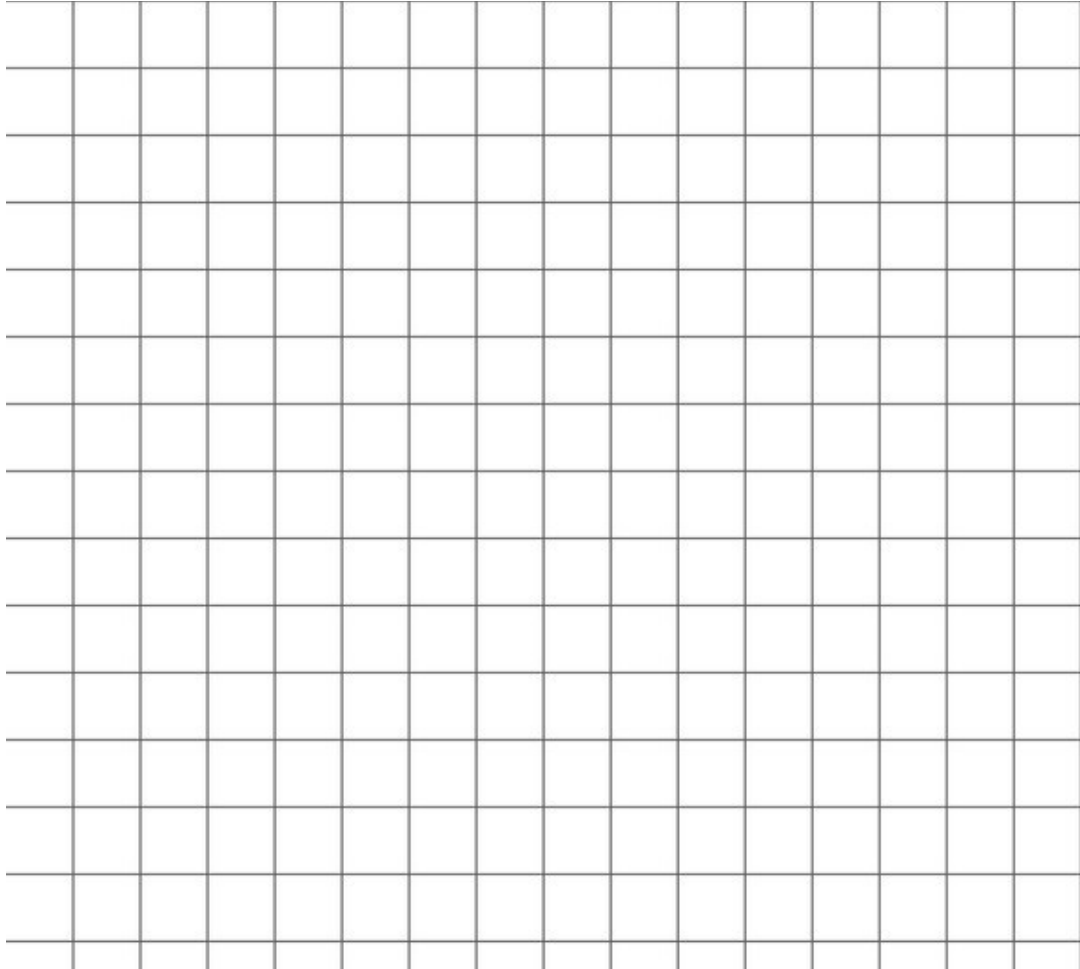
Research Overview and Philosophy



Harnessing Materials Imperfections for Energy Sustainability

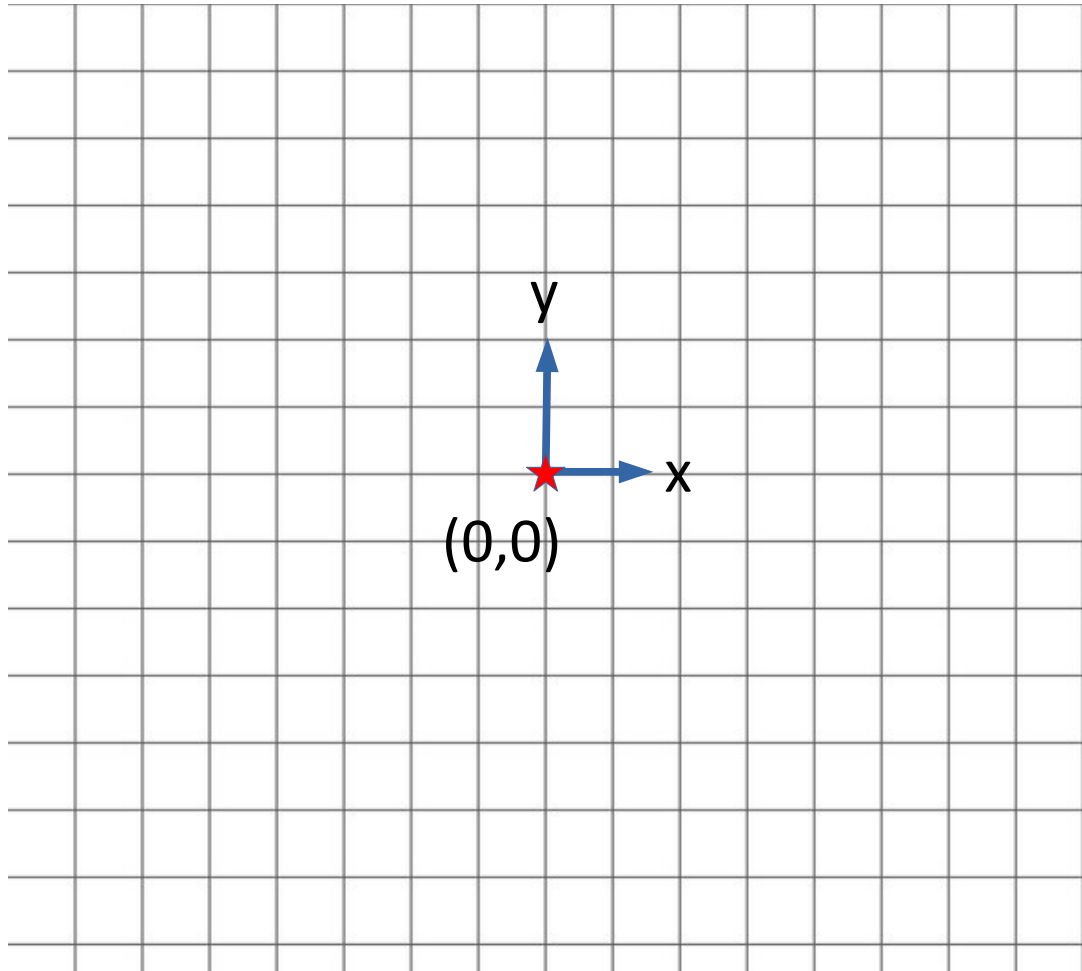


Random walk diffusion: an atomic model for diffusion



Rules for the random walker:

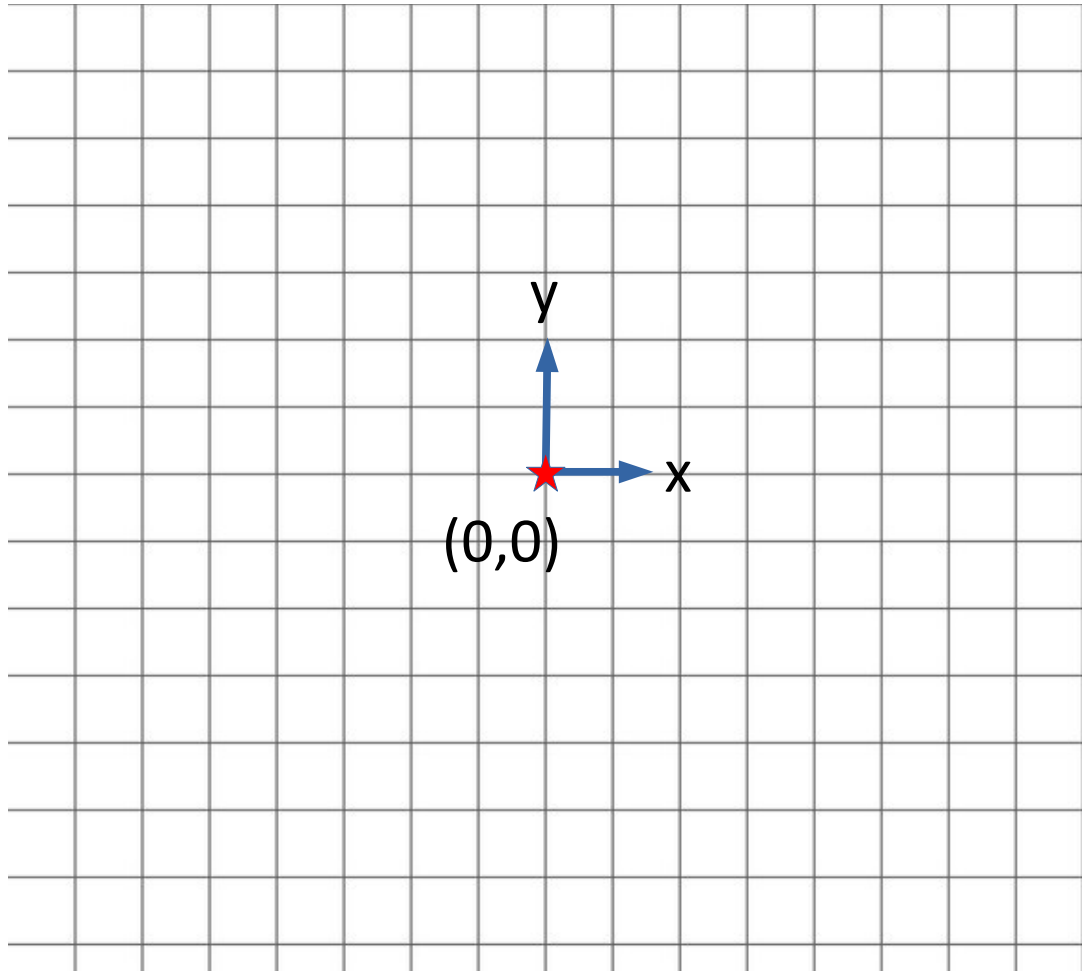
Random walk diffusion: an atomic model for diffusion



Rules for the random walker:

- divide time into nt discrete steps spaced by Δt time, where nt is an integer and Δt is a number
- can only move 1 space at each time step
- equal and random probability of moving up, down, left, right

Random walk diffusion: an atomic model for diffusion



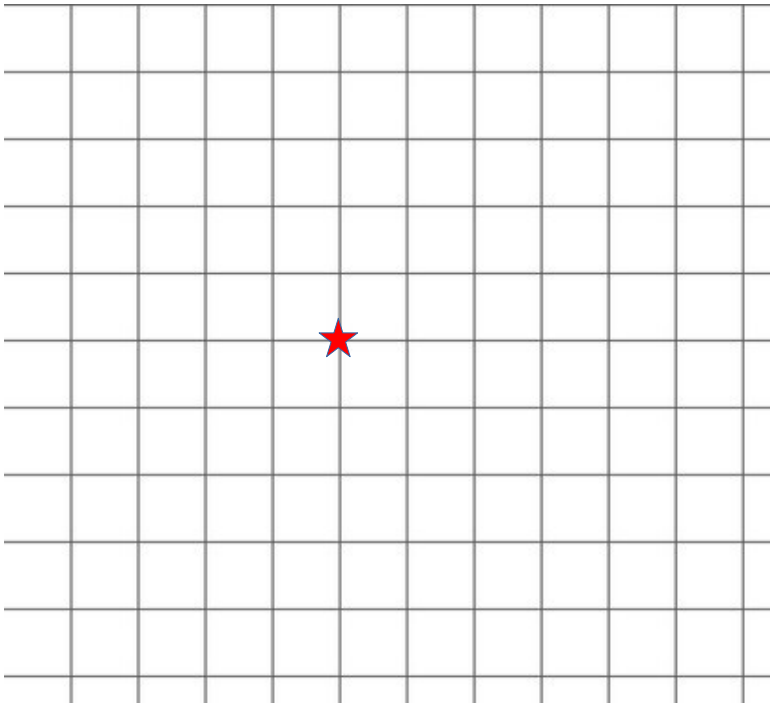
Rules for the random walker:

- divide time into nt discrete steps spaced by Δt time, where nt is an integer and Δt is a number
- can only move 1 space at each time step
- equal and random probability of moving up, down, left, right

If the random walker starts at the red star, where will it be after many ($n \gg 1$) steps?

An implementation of the 2D Random Walk Diffusion

- Example: $nt = 10$ time steps, $\Delta t = 1$ s
Total time is $t = nt * \Delta t = 10$ s



Start at origin $(x,y) = (0,0)$ at time $t = 0$

Use a random number generator and encode the following:

- 1 = move up
- 2 = move down
- 3 = move left
- 4 = move right

Random walk diffusion: a small simulation

<https://rwd2d-mercury.runmercury.com/>



A brief intro to Python

Python is a high-level, general-purpose programming language.

Its design philosophy emphasizes code readability.

This makes Python a great language to learn basic programming concepts, which can be used to learn other programming languages.



A brief intro to Python

Python is a high-level, general-purpose programming language.

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This makes Python a great language to learn basic programming concepts, used to learn other programming languages.



Just like any other language, there are rules on how to write code. We call these rules the ***syntax***.

For example, consider the role of the comma in these two sentences:

Let's eat grandma vs Let's eat, grandma



Computers need *exact* instructions, so poor syntax (e.g., typos) will lead to code errors, Let's learn some programming rules.

Let's look at a Python code

```
vim lattice_2D_EXERCISE.py
1 # Random walk on a 2D square lattice
2
3 # UT MITE 2024 Summer activity
4
5 #-----
6
7 # import relevant libraries,
8 # contains useful functions already implemented
9 import numpy as np
10 import matplotlib.pyplot as plt
11
12 # we define a function that encapsulates all the code related to
13 # generating the path for a random walker
14 def random_walk(nt):
15     """
16     Random walk on a 2D square lattice
17     Inputs:
18         nt [integer] = number of desired jumps (i.e., time steps)
19     Outputs:
20         x [array] = x-coordinate at each time step
21         y [array] = y-coordinate at each time step
22     """
23     # array for x- and y-coordinates along hopping path
24     x = np.zeros(nt+1)
25     y = np.zeros(nt+1)
26     rs2 = np.zeros(nt+1)
27
28 "lattice_2D_EXERCISE.py" 71L, 1986B
```

To make code easier to read for humans, we often put in descriptive **comments**

Two types of comments: line comments (marked with #) and block comments (marked with """)

Let's look at a Python code

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```

Python is **modular**, so you can pick and choose to add different packages, which we will **import**

numpy let's us work more easily with numerical data

matplotlib is for generating plots

Let's look at a Python code (advanced)

```
vim lattice_2D_EXERCISE.py
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26     rs2 = np.zeros(nt+1)
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28     # lattice_2D_EXERCISE.py" 71L, 1986B
```

If a piece of code gets used many times, you can put it in a **function**

Here, we are defining a function called **random_walk** that takes in an input called **nt** to be specified later.

Let's look at a Python code

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```

We will use different types of variables to store different types of numerical data

Let's look at a Python code

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```

We use the **numpy** package to help generate our first variable

An implementation of the 2D Random Walk Diffusion

Objective

User chooses nt = number of time steps

Concept

Variable, integer

Python Representation

`nt`

An implementation of the 2D Random Walk Diffusion

Objective

User chooses nt = number of time steps

Keep track of position of random walker at each time step.
Let's assume it starts at the origin.

Concept

Variable, integer

Array (list of items)
e.g., [3, 4.5, 8, -1]

2D → x and y coordinate
for each position

Python Representation

nt

Use the library numpy,
shorthand is np:

```
x = np.zeros(nt+1)
y = np.zeros(nt+1)
```

An implementation of the 2D Random Walk Diffusion

Objective

User chooses nt = number of time steps

Keep track of position of random walker at each time step.
Let's assume it starts at the origin.

Specify how the position changes at each time step.

Concept

Variable, integer

Array (list of items)
e.g., [3, 4.5, 8, -1]

2D → x and y coordinate
for each position

Python Representation

nt

Use the library numpy,
shorthand is np:

```
x = np.zeros(nt+1)
y = np.zeros(nt+1)
```

```
delx = np.array([?, ?, ?, ?])
dely = np.array([?, ?, ?, ?])
```

An implementation of the 2D Random Walk Diffusion

Objective

User chooses nt = number of time steps

Keep track of position of random walker at each time step.

Let's assume it starts at the origin.

Specify how the position changes at each time step.

Save each new position of the diffusion path

Concept

Variable, integer

Array (list of items)
e.g., [3, 4.5, 8, -1]

2D $\rightarrow$ x and y coordinate
for each position

Index the array
i.e., access a specific element
“zero index”

Python Representation

nt

Use the library numpy,
shorthand is np:

```
x = np.zeros(nt+1)
y = np.zeros(nt+1)
```

```
delx = np.array([?, ?, ?, ?])
dely = np.array([?, ?, ?, ?])
```

```
x = [1, 2, 3]
x[0] = 1
x[1] = 2
```

An implementation of the 2D Random Walk Diffusion

Objective

Repeat for nt times

Encode the random number to a change in position of the random walker

Concept

for loop
range function

Generate a (pseudo)-random number

Python Representation

Input:

```
for i in range(3):  
    print(i)
```

Output:

```
0  
1  
2
```

```
np.floor(4* np.random.rand(nt))
```

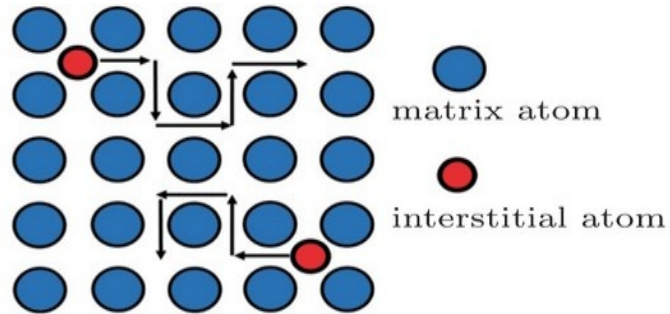
Generate random number b/t 0 and 1

Random number b/t 0 and 4

Random integer: 0, 1, 2, 3

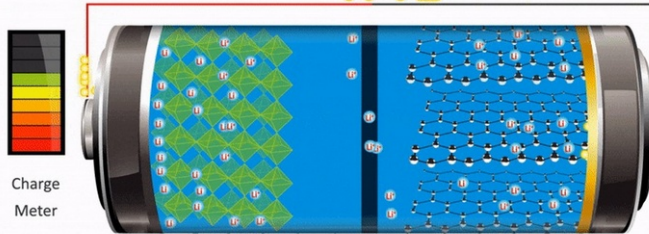
Let's try a little coding

Does the Random Walk Diffusion model have any correspondence with a real material?

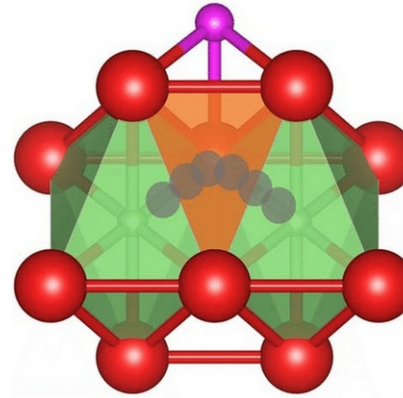


How Lithium-ion Batteries Work

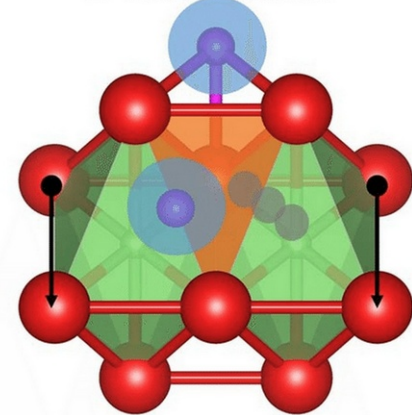
Discharge



1-TM Diffusion



2-TM Diffusion



Florian Schipper et al 2017 J. Electrochem. Soc. 164 A6220

Things learned today

Materials Science and Engineering and the Materials Science Tetrahedron

Diffusion for mass transport

Examples of diffusion

- Soda, air fresheners

- Impurities in semiconductors, Li-ion battery

Computation and simulation of diffusion

- Random Walk Diffusion

More questions? Reach out at <https://wangmaterialsgroup.com>